ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 17	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
2	**(a)** VIII	8 is written as VIII
3	**(a)** 5	Add the powers of 2 where there is a 1
4	**(c)** 101	Divide repeatedly by 2 and read the remainders upwards
5	**(c)** 600	Place value = digit x value of its position
6	**(b)** 0 to 1	Base 2 uses 2 digits starting from 0
7	**(d)** L	Learn the seven Roman symbols
8	**(a)** 24	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
9	**(a)** XIX	19 is written as XIX
10	**(b)** 6	Add the powers of 2 where there is a 1

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	27 = XXVII (2).	2
12	$1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 7$ (1+1).	1+1
13	Repeated division by 2 gives 7 = 111_2 (1+1).	1+1
14	44 = XLIV (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	XLVII = 47 (1); CLXIV = 164 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
16	$4 \times 10^3 + 0 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$ (3).	3
17	$1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0$ (1+1); $= 38$ (1).	1+1+1
	OR: XXXVIII = 38 (1); CCXXIX = 229 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
18	$7 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 5 \times 10^0$ (3).	3
	OR: $2 \times 5^2 + 4 \times 5^1 + 1 \times 5^0$ (1+1); $= 71$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	12 = XII (1); 25 = XXV (1); 38 = XXXVIII (1); 46 = XLVI (1); 59 = LIX (1).	1+1+1+1+1
	OR: Steps: $19 = 2 \times 9 + 1$; $9 = 2 \times 4 + 1$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 10011_2 (2).	3+2
20	14 = XIV (1); 29 = XXIX (1); 43 = XLIII (1); 67 = LXVII (1); 82 = LXXXII (1).	1+1+1+1+1
	OR: Steps: $26 = 2 \times 13 + 0$; $13 = 2 \times 6 + 1$; $6 = 2 \times 3 + 0$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 11010_2 (2).	3+2

Mistakes to avoid:

- Repeating V, L or D, or writing IC, IL, XM for 99, 49, 990; use XCIX, XLIX and CMXC.
- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.

- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1001	Divide repeatedly by 2 and read the remainders upwards
2	**(c)** 400	Place value = digit x value of its position
3	**(d)** 0 to 2	Base 3 uses 3 digits starting from 0
4	**(c)** C	Learn the seven Roman symbols
5	**(a)** 39	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
6	**(a)** XXXIV	34 is written as XXXIV
7	**(b)** 9	Add the powers of 2 where there is a 1
8	**(b)** 1100	Divide repeatedly by 2 and read the remainders upwards
9	**(a)** 700	Place value = digit x value of its position
10	**(d)** 0 to 3	Base 4 uses 4 digits starting from 0

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 10$ (1+1).	1+1
12	Repeated division by 2 gives 10 = 1010_2 (1+1).	1+1
13	78 = LXXVIII (2).	2
14	$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 12$ (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LXXXIV = 84 (1); CCCXC = 390 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
16	$2 \times 10^3 + 9 \times 10^2 + 1 \times 10^1 + 8 \times 10^0$ (3).	3
17	$3 \times 5^2 + 0 \times 5^1 + 4 \times 5^0$ (1+1); $= 79$ (1).	1+1+1
	OR: LIX = 59 (1); CXLVIII = 148 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
18	$6 \times 10^3 + 4 \times 10^2 + 2 \times 10^1 + 0 \times 10^0$ (3).	3
	OR: $4 \times 5^2 + 1 \times 5^1 + 2 \times 5^0$ (1+1); $= 107$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	16 = XVI (1); 33 = XXXIII (1); 48 = XLVIII (1); 74 = LXXIV (1); 99 = XCIX (1).	1+1+1+1+1
	OR: Steps: $35 = 2 \times 17 + 1$; $17 = 2 \times 8 + 1$; $8 = 2 \times 4 + 0$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 100011_2 (2).	3+2
20	18 = XVIII (1); 37 = XXXVII (1); 52 = LII (1); 88 = LXXXVIII (1); 94 = XCIV (1).	1+1+1+1+1
	OR: Steps: $41 = 2 \times 20 + 1$; $20 = 2 \times 10 + 0$; $10 = 2 \times 5 + 0$; $5 = 2 \times 2 + 1$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 101001_2 (2).	3+2

Mistakes to avoid:

- Repeating V, L or D, or writing IC, IL, XM for 99, 49, 990; use XCIX, XLIX and CMXC.
- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** D	Learn the seven Roman symbols
2	**(b)** 56	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
3	**(a)** XLIX	49 is written as XLIX
4	**(c)** 11	Add the powers of 2 where there is a 1
5	**(a)** 1101	Divide repeatedly by 2 and read the remainders upwards
6	**(c)** 9000	Place value = digit x value of its position
7	**(c)** 0 to 4	Base 5 uses 5 digits starting from 0
8	**(c)** M	Learn the seven Roman symbols
9	**(d)** 90	Add the values of the symbols, subtracting a smaller symbol placed before a larger one
10	**(b)** LXVI	66 is written as LXVI

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Repeated division by 2 gives 15 = 1111_2 (1+1).	1+1
12	95 = XCV (2).	2
13	$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 17$ (1+1).	1+1
14	Repeated division by 2 gives 18 = 10010_2 (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LXXIII = 73 (1); CCXVI = 216 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
16	$9 \times 10^3 + 0 \times 10^2 + 5 \times 10^1 + 7 \times 10^0$ (3).	3
17	$1 \times 5^2 + 3 \times 5^1 + 4 \times 5^0$ (1+1); $= 44$ (1).	1+1+1
	OR: XCVI = 96 (1); CDLXXV = 475 (1); add the symbol values, subtract when a smaller symbol comes before a larger one (1).	1+1+1
18	$3 \times 10^3 + 7 \times 10^2 + 6 \times 10^1 + 4 \times 10^0$ (3).	3
	OR: $2 \times 5^2 + 3 \times 5^1 + 0 \times 5^0$ (1+1); $= 65$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	21 = XXI (1); 45 = XLV (1); 63 = LXIII (1); 79 = LXXIX (1); 97 = XCVII (1).	1+1+1+1+1
	OR: Steps: $52 = 2 \times 26 + 0$; $26 = 2 \times 13 + 0$; $13 = 2 \times 6 + 1$; $6 = 2 \times 3 + 0$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 110100_2 (2).	3+2
20	24 = XXIV (1); 39 = XXXIX (1); 56 = LVI (1); 71 = LXXI (1); 86 = LXXXVI (1).	1+1+1+1+1
	OR: Steps: $58 = 2 \times 29 + 0$; $29 = 2 \times 14 + 1$; $14 = 2 \times 7 + 0$; $7 = 2 \times 3 + 1$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (3); reading remainders from last to first gives 111010_2 (2).	3+2

Mistakes to avoid:

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- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

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